MILITARY SPECIFICATION
CARRIER, CARGO, 1 1/2 TON, M973, SMALL UNIT SUPPORT VEHICLE (SUSV)

This specification is approved for use by Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of full-tracked diesel powered vehicle designed for use as a Small Unit Support Vehicle (SUSV).

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and Standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.
VV-L-800	- Lubricating Oil, General Purpose, Preservative, (Water-displacing, Low Temperature).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2320

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MIL-T-704	- Treatment and Painting of Material.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-L-3150	- Lubricating Oil, Preservative, Medium.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordinance.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-46152	- Lubricating Oil, Internal Combustion Engine, Administrative Service.
MIL-R-0046164	- Rustproofing for New and Fielded Military Vehicles.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone.
MIL-T-62314	- Test Equipment (Simplified) for Internal Combustion Engine (STE/ICE).

STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristic, Measurement of.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-882	- System Safety Program Requirements.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1180	- Safety Standards for Military Ground Vehicles.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and facilities.
MIL-STD-1474	- Noise Limits for Army Material.
MS 35000	- Battery, Storage, Lead-Acid, Waterproof
MS 53028	- Chassis, Trailer: 3/4 Ton, 2 Wheel, Military Design, M116A1
MS 53085	- Trailer, Cargo, 3/4 Ton, 2 Wheel, M101A1 (Military Design)

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MS 75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.
MS 75021-2	- Connector, Receptacle, Electrical - 12 Contact,Intervehicular, 28 Volt, Waterproof.

2.1.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AR 70-38	- Research, Development, Test and Evaluation of Materiel for Extreme Climatic Conditions.
TB-MED-501	- Hearing Conservation.
NATO STANAG 2040	- Stretchers (Edition No. 4) Release to Other Than DoD Agencies must be Approved By HQ Darcom (DRCID).
NATO STANAG 4074	- Standard Nato Slave Connector for Auxiliary Power.

DRAWINGS

ARMY

7041347	- Lunette, Coupler Drawbar,Ring.
7731428	- Cover Assembly, Receptacle.
10916537	- Fire Extinguisher.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article (preproduction). The contractor shall furnish sample units for first article inspection and approval (see 4.3 and 6.3). First article samples shall be inspected by the contractor under quality assurance provisions of this specification. First article samples shall be fully representative of units to be supplied from production tooling and facilities.

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3.2 Initial production vehicles. Vehicles manufactured at initiation of quantity production shall be inspected and shall pass initial production inspection (see 4.4). Initial production vehicles shall be representative of vehicles to be furnished under this specification and shall include all current modifications. Modification shall not be applied to subsequent vehicles without approval of the Government.

3.2.1 Reliability. The reliability of the vehicle(s) shall not be less than 330 Mean Miles between failure (MMBF) (531 Mean Kilometers Between Failure (MKBF)) during the first 2500 miles, when operated with the operational profile (see 3.2.1.1).

3.2.1.1 Operational profile. Test mileage shall be apportioned as follows:

a.	Hard-surfaced roads	10 percent
b.	Secondary roads	30 percent
c.	Cross-country roads	60 percent

3.2.1.2 Failure. A failure is a malfunction of an end item requiring overhaul, replacement, rebuild salvage, or other corrective action as prescribed by maintenance manual which cannot be deferred for remainder of miles.

3.2.1.3 Corrective action. Corrective action is not deferrable if malfunction causes:

a. inability to commence operation or cessation of operation.
b. inability to conform to requirements of specification.
c. critical or catastrophic hazard as defined in MIL-STD-882.

3.2.1.4 Malfunction. Any malfunction which operator or crew can remedy shall not be considered a failure provided that repair is authorized or prescribed as an operator function and can be accomplished within 15 minutes using only controls, tools, or spare parts incorporated or carried with end item or its prime mover.

3.2.2 Maintainability. Total maintenance, excluding driver and crew checks and services, shall not exceed 72.5 manhours during 2500 miles of operation in accordance with the operational profile (see 3.2.1.1). This equates to a maintenance ratio (MR) of 0.29 at 10 operational miles equivalent to one operational hour. Scheduled maintenance interval shall be 3 months or 2500 miles, whichever comes first. Mean time to repair (maintenance clock hours divided by number of unscheduled maintenance actions) for SUSV hardware, shall not exceed the following levels for the indicated repair levels.

a. Organizational maintenance level shall not exceed 2 manhours 92 percent of the time.
b. Direct support maintenance level shall not exceed 8 manhours 92 percent of the time.

3.2.3 Transportability. Vehicle shall be capable of being lifted at GVW (see 3.4.1.5) by a CH-47C helicopter using standard U.S. Army slings. Use of

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sling lift spreader bars is not encouraged. Vehicle shall be capable of being transported internally by C-130 type aircraft, rail, M915 series truck tractor with appropriate semi-trailer, or surface vessel in a complete configuration with components stowed. Vehicle shall incorporate hard points for sling lift and aircraft tiedown. Hard points for sling lift or aircraft tiedown shall comply with requirements of MIL-STD-209. A durable and waterproof diagram with instructions for lifting and tiedown arrangements shall be provided with each vehicle.

3.3 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free from defects which adversely affect performance or serviceability of the finished product (see 6.6). Materials other than those specified shall be of accepted commercial quality.

3.3.1 Qualified products. Contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPL's) whenever available. Contractor inspection records shall specifically °list all QPL items by number and date of the QPL, name of supplier, and part or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, contractor shall list products by number and date of approved document and name of supplier(s).

3.3.2 Component ratings. All system components shall be rated by component manufacturer for vehicle application. Ratings shall not be raised to meet requirements of this specification.

3.3.3 Certification. The supplier and his major component suppliers shall certify that chassis, components, and materials conform to requirements specified herein.

3.3.4 Ozone resistance. When rubber components are required to be ozone resistant, degree of resistance shall be as specified in applicable specifications or drawings.

3.3.5 Dissimilar metals. Dissimilar metals shall be protected from galvanic corrosion in accordance with requirements of MIL-STD-889.

3.3.6 Standard parts. Military standard parts shall be incorporated wherever possible. Commercial standard parts may be used provided they are interchangeable with military standard parts without modification.

3.3.7 Locking devices. Lock washers, self-locking nuts, safety wires, or other approved locking devices shall be incorporated where specified or where required to prevent loosening of components.

3.4 Design and construction

3.4.1 Physical characteristics.

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3.4.1.1 Physical configuration. Vehicle shall be a single unit or articulated two-unit configuration.

3.4.1.2 Track. Vehicle shall be full tracked and shall use one type track for year-round service on all terrain.

3.4.1.3 Curb weight. Curb weight of vehicle shall not exceed 10,500 pounds (4762.8 kilograms (kg)). Curb weight shall include weight of vehicle with all attachments, accessories, equipment, and a full complement of fuels lubricants, and coolant.

3.4.1.4 Payload. Vehicle shall have a payload capacity of not less than 3,000 pounds (1360 kg). Vehicle payload shall include driver, personnel and gear, cargo, basic issue items and kits. Payload shall be evenly distributed over vehicle body load space area.

3.4.1.5 Gross vehicle weight (GVW). GVW shall be the sum of curb weight (see 3.4.1.3) and payload (see 3.4.1.4).

3.4.1.6 Ground pressure. Vehicle at GVW shall have a ground pressure not exceeding 2.5 pounds force per square inch (psi) (17.24 kilopascals (kPa)).

3.4.1.7 Cargo space. Cargo space of the vehicle shall be enclosed and shall be a minimum length of 98 inches (248.92 cm), minimum width of 64 inches (162.0 cm), and a minimum height of 49 inches (124.0 cm) excluding handles attached to inside of roof.

3.4.1.8 Roof hatch. Vehicle shall have at least one roof hatch with cover in driver's compartment. Hatch opening size shall be at least 22 inches by 22 inches (559 mm by 559 mm).

3.4.1.9 Ground clearance. Vehicle shall have a ground clearance not less than 12 inches (30.48 cm) when measured on a hard, flat surface.

3.4.1.10 Mounts, mounting devices and brackets. Vehicle shall be capable of accepting mounts, mounting devices, and brackets to:

a. Accommodate 4 litters, 90 inches (228.6cm) long, 24 inches (60.96cm) wide, and 6 inches (15.24cm) high (STANAG 2040), each weighing approximately 10 lbs (4.54kg) when used in a MEDEVAC role.

b. Stow two portable 2 3/4 pound (1.14 kg) liquid CF_3BR fire extinguishers conforming to Drawing 10916537 within compartments. One of the extinguishers shall be readily accessible to the crew.

c. Accommodate portable decontamination apparatus (M11 or equivalent).

d. Accommodate and power an automatic chemical agent detector or alarm system (M43A1/M42). Space shall be provided to store a chemical agent detector kit.

e. Accommodate the US M16A1 rifles of driver and assistant driver.

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f. Accommodate and power installation kits for the following standard U.S. Army radios, radio mounts, and security devices:

(1) AN/VRC46
(2) AN/VRC47
(3) AN/VRC48
(4) AN/VRC49
(5) AN/GRC160
(6) TSEC/KY57 or TSEC/KY58

g. Accommodate and power a radiac detector/alarm system (AN/VDR-2)

h. Tow two 10 men squads on skis.

i. Tow sled, trailer, or howitzer (see 3.5.15). Tow pintle assembly shall be capable of accommodating the Lunette-coupler drawbar ring (Drawing 7041347).

j. Accommodate standard vehicle log book.

k. Accommodate standard vehicle first aid kit.

l. Accommodate standard vehicle slave starting connector.

m. Accommodate standard vehilce tow cables.

3.4.1.11 Crew and cargo protection. Crew and cargo compartment inclosure(s) shall afford protection from exposure of liquid chemical agent contamination.

3.4.1.12 Engine/fuel. Vehicle shall have fuel diesel engine capable of operating on fuel conforming to all grades of VV-F-800. A water separator of fuel filter permitting drainage of condensation from filter before daily operation shall be provided.

3.4.1.13 Transmission. An automatic transmission is required. Transmission shift lever location shall be readily accessible to driver and shall be capable of being operated by a 5th thru 95th percentile soldier wearing arctic climatic clothing.

3.4.1.14 Lubricants. Vehicle shall be operable using standard military lubricants without adverse effect on vehicle components. Lubrication intervals shall be as required by component manufacturers. A lubrication chart shall be furnished with each vehicle and shall specify interchangeable military lubricants as well as commercial lubricants. All initial fills of lubricants shall conform to MIL-L-46152, MIL-L-2105 and MIL-L-46167.

3.4.1.15 Electrical system. Vehicle shall be equipped with a 24 volt direct current (vDC) electrical system complete with necessary items of equipment, wiring and electrical plugs to power on-board equipment, and towed equipment.

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3.4.1.15.1 Electrical connectors. Electrical plugs shall be located and protected to assure wiring and plugs shall not be crushed, chafed, burned, or otherwise damaged during operation.

3.4.1.15.2 Electrical circuits. Electrical circuits shall maintain continuity from end to end, without evidence of internal or external shorts during all vehicle operating conditions. All electrical circuits will be electrically protected with appropriate electrical fuses and circuit breakers.

3.4.1.15.3 Batteries. Vehicle shall be equipped with two 2 HN type batteries, each with a minimum capacity of 100 ampere-hours conforming to MS35000. Batteries shall have their negative terminals ground and both battery terminals identified.

TABLE I. Receptacle terminal connections.

Terminal	To provide
Terminal A	LH Blackout Marker
Terminal B	LH Turn Signal and Stop Lights
Terminal C	RH Blackout Marker
Terminal D	Ground
Terminal E	Tail light on right and Left Rear
Terminal F	Blackout Stop
Terminal H	Blackout Clearance
Terminal J	RH Turn Signal and Stop Lights

3.4.1.15.4 Lighting and blackout. Vehicle shall be equipped with service and blackout lighting system. Spectral emission of lamps used in blackout conditions shall be such that no energy will be emitted in 700 to 1200 nanometer spectrum. Lamps used to provide illumination for crew visual requirements will provide peak emission in the 450 to 500 nanometer spectrum. Vehicle shall be equipped with blackout window shades in cargo compartment(s).

3.4.1.15.5 Trailer lighting receptacle. Provision shall be made for trailer lighting wiring harness MS 75021-2 receptacle or equivalent and cover asssembly in accordance with Drawing 7731428. Receptacle and cover assembly shall be located to provide clearance as detailed in MS 75020, MS 53028, and MS 53085. Receptacle shall be wired into vehicle electrical system as specified in Table I and MS 75021-2.

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3.4.1.16 Intercommunication device signal. If driver compartment is separated from cargo space to preclude voice communication, a device shall be provided to facilitate signaling driver from cargo space. Device shall include a signal light and audio device in driver compartment and be capable of being operated from cargo space.

3.4.1.17 Accessories and equipment. Vehicle shall have all standard accessories normally furnished with vehicle and include the following equipment:

a. Ignition switch.
b. Ammeter warning light.
c. Fuel gage.
d. Oil pressure light.
e. Engine temperature gage.
f. Speedometer.
g. Recording odometer.
h. Headlight light beam indicator light.
i. Driver rear view mirrors on both sides of vehicle body. Exterior mirrors shall be capable of folding toward body in at least one direction.
j. Interior lights.
k. Horn with suitable electromagnetic.
l. Manually operated valve(s) on engine for purpose of withdrawing oil samples for analysis.
m. Lighted automatic transmission indicator.

3.4.1.18 Heaters.

3.4.1.18.1 Personnel heater. Personnel heaters shall be of sufficient capability and capacity to assure that cab reference temperature at waist level in driver area of 41 degrees Fahrenheit(^{o}F) 5 degrees Celsius(^{o}C) is attained within 45 minutes after heater is turned on while vehicle is being driven and when ambient air temperature is minus 50°F (-46°C). A windshield defroster shall be provided for defrosting windshield and windows to left and right of driver when ambient air temperature ranges from 95°F to minus 50°F (35°C to -46°C). Forced fresh air (outside air) ventilation system shall be provided for summer use. Heater(s) in cargo space, shall be of sufficient capability and capacity to assure that cargo space reference temperature of 35°F (1.8^{o}C) at waist level is attained within 45 minutes after heater is turned on when ambient air temperature is minus 50°F (-46°C) and while vehicle is being driven.

3.4.1.18.2 Engine heater. The engine heater system shall be an on-board system capable of bringing engine to a temperature at which it can be started with 30 seconds crank time after 45 minutes of heat application after vehicle has been cold soaked at minus 50°F (-46°C) for 24 hours. Vehicle electrical batteries shall not be the source of energy used for heating engine.

3.4.1.18.3 Battery heater. An on-board system shall be capable of maintaining battery temperature to allow a full charge when engine is operating. Vehicle electrical batteries shall not be the source of energy used for heating batteries.

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3.4.1.19 Windshield wipers and washers. Vehicle shall be equipped with electrically operated windshield wipers and washers in accordance with Requirement 104 of MIL-STD-1180 and shall be examined for manual operation.

3.4.1.20 Bilge pump(s). Vehicle shall be equipped with bilge pump(s) installed in accordance with applicable drawings. Bilge pumps shall remove water from the vehicle at a rate of 36 gallons per minute (136 liters per minute) with a discharge head of 3 feet (91.44cm) and with pump motor operating on 24 vdc impressed at the motor. Pump(s) shall be installed at lowest point in each unit of vehicle body and shall be capable of operating from the driver's position. Bilge pump discharge outlet should be at least 6 inches (15.24cm) above water line when floating at GVW in calm fresh water.

3.4.1.21 Roof and roof rack. The roof of the vehicle shall support a 250 pound (113kg) man and shall have an anti-skid coating. Roof rack shall provide for strapping such items as 210 centimeter(cm) (82.68 inch) skis and snowshoes to the rack. Built-in steps or ladders shall be provided to allow a soldier in arctic clothing to stow and strap down 36 (18 pairs) 210 cm long skis to the rack within 10 minutes.

3.4.1.22 Power winch. Vehicle shall be equipped with power winch deployed from front of vehicle. Winch shall be capable of pulling 6,500 lbs (2948kg) on bare drum (first layer) with 82 ft (25.0 m) of cable installed. Cable shall have a minimum breaking load of 13,000 lbs (5879 kg). The winch system shall incorporate features to prevent winch and cable overloading. Winch loading shall be limited to 110 percent of winch rated capacity if hydraulic protective-devices (relief valves) are used.

3.4.1.23 Chemical biological (CB) clothing storage container. Storage container for driver and co-driver CB protective clothing shall be provided. Minimum storage container size for one crew member CB protective clothing shall be at least 12 inches (30.48 cm) long, 18 inches (45.12 cm) wide, and 6 inches (15.24 cm) deep.

3.5 Performance. Vehicle shall meet performance requirements of this specification when equipped as specified herein and serviced with the standard service products specified in table II.

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TABLE II. Service products specifications.

Product use	Ambient air temperatures -65 to 0°F	-10° to +132°F
Oil:		
For hydraulic brakes	MIL-B-46176	MIL-B-46176
For shock absorbers	MIL-H-5606	MIL-H-5606
For general purpose lubrication	VV-L-800	MIL-L-3150
Grease:		
For sealed bearings	MIL-G-23827	MIL-G-23827
For general chassis lubrication, including wheel bearings	MIL-G-10924	MIL-G-10924

3.5.1 Speed. On level hard surfaced roads, vehicle at GVW (see 3.4.1.5) shall be capable of maintaining a top speed of at least 25 mph (40.23 kmph) and a speed of 2.5 mph (4.02 kmph) or lower for thirty minutes in ambient air temperature of 95°F (35°C) without overheating, steering difficulties, or malfunction of any component.

3.5.2 Terrain mix. Vehicle at GVW shall be operated over the operational profile (see 3.2.1.1) during all seasons and shall be capable of continuous operation during terrain changes (from road to cross-country and vice versa) without modification. Vehicle operating on the terrain mix based on a 24 hour operating day, shall conform to table III.

TABLE III. Vehicle movement.

Movement rate	Time-hours	Distance traveled-miles (km) Minimum	Maximum
Parked, engine running	7.20	0	0
Parked, engine stopped	14.40	0	0
3-8 mph (4.83-12.87 kmph)	1.44	4.32(6.95)	11.52(18.54)
8-20 mph (32.18 kmph)	0.48	9.60(15.45)	9.60(15.45)
Over 20 mph (32.18 kmph)	0.48	9.60(15.45)	9.60.(15.45)
TOTAL	24.00	17.76 (28.58)	30.72 (49.44)

3.5.3 Winter operation. Vehicle at GVW shall be capable of operating on cross-country terrain covered with snow up to 6 feet (1.83m) deep. Vehicle shall be capable of operating over snow-covered powder and hard-packed roads.

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When operating at GVW and towing two 10 men infantry squads on skis, vehicle shall negotiate terrain mix of 3.5.2, (when covered by snow at temperatures not exceeding 14°F (-10°C)) and shall traverse at least 2.2 miles (3.5 km) in 1-hour.

3.5.4 Spring break-up and summer operation. Vehicle at GVW shall be capable of crossing muskeg bogs. Vehicle shall be capable of operating on muddy, gravel, and stone-covered roads and shall negotiate the terrain mix of 3.5.2 during spring break-up and summer, while traversing at least 4.4 miles (7.0km) in one hour.

3.5.5 Gradeability. Vehicle at GVW and without modification, shall be capable of ascending and descending forward and backward on hard, dry longitudinal grades 60 percent at a sustained speed of not less than 2 mph (3.22 kmph). Vehicle shall be capable of stopping and holding motionless on 60 percent grade, restarting and continuing to ascend and descend the 60 percent grade. At a sustained speed of not less than 2mph (3.22 kmph), vehicle shall be capable of ascending and descending up to 30 percent longitudinal grades covered with not less than 1.5 feet (0.46m) of loose, dry snow.

3.5.6 Side slopes. Vehicle at GVW and without modification shall be capable of traversing a 40 percent hard, dry side slope without slipping sideways with both the left and right sides up slope. The steering system must be responsive and effective. Vehicle shall be capable of stopping and holding motionless on the 40 percent side slope, restarting and continuing to traverse the 40 percent side slope. Vehicle shall be capable of operating on the side slope without malfunction, leakage, or damage to vehicle. Vehicle shall be capable of traversing a 30 percent side slope covered with not less than 1.5 feet (0.46m) of loose, dry snow.

3.5.7 Acceleration. Vehicle at GVW shall be capable of accelerating on dry, hard level roads from 0 to 25 mph (0 to 40.23 kmph) in less than 20 seconds.

3.5.8 Service brakes. Service brakes shall be capable of controlling and holding vehicle at GVW headed uphill or downgrade on a 60 percent grade. Under all conditions of loading, service brakes shall be capable of stopping vehicle from a speed of 20 mph (32.18 kmph) in 35 feet (10.67m) or less from point of brake application. Under all conditions of loading, service brakes shall be capable of stopping vehicle when towing a fully loaded 1 1/2 ton trailer from a speed of 20 mph (32.18 kmph) in 50 feet (15.24m) or less from point of brake application. While braking, vehicle with or without towed load must remain in a 12 foot (3.66m) wide roadway during all stopping actions. Stopping distance shall be demonstrated on a dry, hard level road surface free from loose material. Brake pedal forces required to achieve stopping distances shall not exceed 100 pounds force (444.8N).

3.5.9 Park brake. With engine running or shut-down and vehicle at GVW, headed either up or downgrade, the parking brake shall be capable of holding vehicle motionless on a hard, dry 40 percent grade free from loose material. Force required to set and release brakes under these conditions shall not exceed 100 pounds force (444.8N).

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3.5.10 Turning ability. Vehicle at GVW shall be capable of sustaining from 0.4 to 0.6g lateral acceleration in constant radius turn. Curb to curb turning circle radius shall not exceed 30 feet (9.14m).

3.5.11 Vertical step. Vehicle at GVW shall be capable of negotiating a 12 inch (30.48cm) vertical step.

3.5.12 Standing vegetation. Vehicle at GVW shall be capable of overcoming light and medium standing vegetation including trees up to 3 inches (7.62 cm) in diameter.

3.5.13 Floatability. Vehicle at GVW shall be capable of floating without vehicle preparation for a minimum of 60 minutes in fresh water with a minimum 6 inch (15.24 cm) freeboard. Following floatation, vehicle shall be capable of performing without malfunction, leakage or damage to vehicle. Vehicle shall be capable of entering and departing both forward and backward from and to a 20 percent prepared dirt slope to and from fresh water of sufficient depth to float vehicle.

3.5.14 Skijoring. Vehicle at GVW shall be capable of towing two 10 men squads on skis over the terrain mix of 3.5.2.

3.5.15 Towing. Vehicle shall be capable of towing a M14A1 sled, a 1/4 ton to 1 1/2 ton trailer or a M101A1 Howitzer (with wheels or wheel and ski combination), over cross-country terrain covered with a minimum of 1.5 feet (0.46m) of loose dry snow.

3.5.16 Range. Vehicle at GVW shall be capable of operation utilizing only fuel contained in on-board permanent fuel tanks for a distance (range) of 180 miles (290.0 km) over improved roads at varying speeds up to the maximum safe speed of 25 mph (40 kmph).

3.5.17 Low temperature. Vehicle shall be capable of being utilized, transported, and stored in conformance with the variable high humidity, basic cold, and cold (B2, C1, and C2) daily cycles defined in AR 70-38. Vehicle shall be capable of being started without use of kits or external aids within 2 minutes crank time at minus 25°F (-32°C) in the basic cold daily cycle. Vehicle shall be capable of being started at minus 50°F (-46°C) in the cold climatic daily cycle with use of on-board kits and the engine heater system (see 3.4.1.18.2). Supplemental fuels shall not be used at any time to start engine.

3.5.18 Power package cooling. The power package cooling of the vehicle shall be accomplished under conditions of 0.7 tractive effort to vehicle weight and at full engine output power on a 95°F (35°C) ambient day. Steady state engine oil and transmission oil temperatures shall not exceed 270°F (132°C) and 300°F (149°C) respectively.

3.6 Electromagnetic interference. Electromagnetic interference shall be controlled as specified for class A3, MIL-STD-461.

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3.7 Painting. Exterior surface coatings shall provide sufficient adhesion for subsequent recoating with other colors conforming to MIL-C-46168. Vehicle shall be painted with chemical agent resistant coating and paint types F and G of MIL-T-704 to facilitate chemical agent decontamination per TM 3-220.

3.8 Rustproofing. Sheet metal areas of vehicle including frame and structural members shall be rustproofed in accordance with MIL-R-0046164 to minimize corrosion during life of vehicle. Materials must allow retreatment or touch up at times of vehicle overhaul or repair.

3.9 Marking. Vehicle exterior marking shall be in accordance with MIL-STD-642. Parts and components shall be marked in accordance with MIL-STD-130. Markings shall be lusterless black in accordance with MIL-C-46168. Instruction, caution, identification, operation, and data plates in English language shall be provided and installed in accordance with manufacturer normal commercial practice. Military model number, nomenclature, National Stock Number (NSN) contract number, date of manufacture, manufacturer serial number, and US Army Registration number shall be embedded or embossed on an additional metal identification plate and installed in the vehicle crew area readily visible location. The chassis shall be equipped with English language instructions and plates or diagrams including procedures to be followed in assembling, operating, or servicing chassis.

3.10 Human engineering. The design shall provide an environment effective procedures, work patterns, and personnel safety minimizing discomfort, distraction, and other factors that hinder human performance. Design shall minimize personnel and training requirements within limits of time, cost and performance trade-offs as in MIL-STD-1472. Crew and passenger stations and entrance and exit provisions shall accommodate a range between a 90th percentile male and a 5th percentile female soldier in arctic clothing.

3.11 Safety. Exposed components and systems which are subject to high temperatures and pressures or could be electrically actuated or inherently hazardous, shall be provided with safeguarding and insulating features. Roll over protection for the crew and passengers, consistent with vehicle operation in its defined mission profile, is required. Ignition locks shall be keyed alike and shall have identical keys. Vehicles shall comply with the following requirements of MIL-STD-1180:

(a) Requirement No. 101 - Motor Vehicle Control Accessibility, Identification and Illumination.

(b) Requirement No. 102 - Transmission Shift Lever Sequence, Starter Interlock and Transmission Braking Effect.

(c) Requirement No. 103 - Windshield Defrosting and Defogging Systems.

(d) Requirement No. 104 - Windshield Wiping and Washing Systems.

(e) Requirement No. 107 - Reflective Surfaces for Components in the Driver Field of View.

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(f) Requirement No. 108 - Lamps, Reflective Devices, and Associated Equipment.

(g) Requirement No. 111 - Rearview Mirrors.

(h) Requirement No. 124 - Accelerator Control Systems.

(i) Requirement No. 125 - Warning Devices.

(j) Requirement No. 205 - Glazing Materials.

(k) Requirement No. 206 - Door Locks and Door Retention Components.

(l) Requirement No. 207 - Seating Systems.

(m) Requirement No. 208 - Occupant Crash Protection.

(n) Requirement No. 209 - Seat Belt Assemblies.

(o) Requirement No. 210 - Seat Belt Assembly Anchorages.

(p) Requirement No. 301 - Fuel System Integrity.

(q) Requirement No. 302 - Flammability of Interior Materials.

3.11.1 Hearing protection. Interior noise levels in personnel occupied areas where hearing protection is not required, shall not exceed limits specified in category D of MIL-STD-1474. If steady state interior noise levels exceed limits set in category D but less than category A as specified in MIL-STD-1474 noise hazard caution signs shall be posted in accordance with MIL-STD-1474 to alert crewmen that hearing protection (TB-MED-501) is required. Vehicle with steady state interior noise levels exceeding category A will not be acceptable.

3.11.2 Toxic gas. Vehicle shall be designed to insure that maximum concentrations of toxic gases in crew and passenger and cargo compartment(s) do not exceed values of Table IV for all operating conditions.

TABLE IV. Maximum concentration levels of toxic gases

GASES	CONCENTRATION-PPM
CO	50
NH3	50
NO2	5
SO2	5

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3.12 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies, shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the standards of industry. All components used in the assembly of the vehicle shall be new.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, contractor may use his own or any other facilities suitable for performance of inspection requirements specified herein, unless disapproved by the Government. Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and service comform to prescribed requirements.

4.1.1 Materials and qualified products. Contractor shall provide Government with evidence that materials (see 3.3) used are as specified in applicable specifications and drawings. Contractor inspection records shall be reviewed to determine conformance to 3.3.1 through 3.3.7.

4.1.2 Parts and components. When requested, contractor shall make available to the Government at time of Government inspection, legible drawings and printed specifications to the end item or any of its manufactured parts and components. Drawings and specifications shall be annotated to reflect latest incorporated revision. Upon completion inspection(s) by the Government, all drawings and specifications will be returned to contractor. Government reserves the right to inspect end items and any parts and components during all manufacturing processed and reject such material that does not conform to either Government or contractor drawings and specifications. All deficiencies detected during any contractor or Government inspection shall be corrected by contractor at no additional cost to the Government. A deficiency is defined as a failure to meet a contract requirement.

4.2 Classification of inspection. Classification of inspection shall be as follows:

a. First article (preproduction vehicle) inspection (see 4.3).
b. First article (initial production vehicle) inspection (see 4.4).
c. First production vehicle inspection (see 4.5).
d. Quality conformance inspection (see 4.6).

4.3 First article (preproduction) inspection. When specified (see 6.3), one preproduction vehicle sample (see 3.1) shall be inspected at a location approved by the Government to determine conformance to requirments of this specification. Inspection shall consist of examination as specified in 4.5.2 and all tests specified herein.

4.3.1 Failure. Failure to pass any examination or test or defects in excess of acceptable quality levels (AQL's) specified, shall be cause for refusal to grant first article approval.

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4.4 First article (initial production vehicles) inspection. Unless otherwise specified (see 6.2), two vehicles from the first month production or the first 20 produced, whichever occurs first, will be subjected to the initial production test.

4.4.1 Initial production test. To determine conformance to Section 3, vehicles will be examined as specified in 4.6.1.3 and 4.6.1.4, and tested as specified in Table VI. Subsequently, both vehicles will be road tested for 2500 miles, as specified in Table V. Inspection will be performed by the Government and will require no more than 90 days. Contractor shall expeditiously furnish repair parts as required to support testing at no cost to the Government.

TABLE V. Sequence of 2500 mile (4025 km) test.

Terrain	Mileage (km)	% of total miles
Improved Roads	250 (402)1/	10
Unimproved Roads	750 (1208)1/	30
Cross-Country Terrain	1500 (2414)1/	60

1/ Test shall be run at varying speeds up to maximum safe speed.

4.4.2 Reliability verification. To determine conformance to 3.2.1, reliability shall be verified at a 60 percent confidence level while vehicles are subjected to 2500 mile initial production road test (see 4.4.1)

4.4.3 Maintainability verification. To determine conformance to 3.2.2, maintainability shall be verified during 2500 mile initial production road test (see 4.4.1). Maintenance shall be performed by the Government at Government approved test site.

4.4.4 Failure. Failure of vehicles to conform to requirements during, or after 2500 mile initial production road test, shall be cause for rejection of vehicles. The Government may refuse to accept further production vehicles until evidence is provided by the contractor that corrective action has been taken. Defects found during or after 2500 mile road test, shall be evidence that vehicles previously accepted are similarly defective unless information to the contrary if provided by the contractor and approved by the Government. All defects shall be corrected on all vehicles by the contractor at no cost to the Goverment.
4.4.2

4.5 First production vehicle inspection.

4.5.1 In-process examination. When specified (see 6.2), during fabrication of the first production vehicle, an in-process examination will be conducted by representatives of TACOM Product Assurance Directorate to evaluate conformance of materials and workmanship to specified requirements. Examination will be made at the contractor or subcontractor facility prior to

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application of primer and paint. Processing and welding procedures, quality system, and inspection records will be evaluated during this examination.

4.5.2 Completed first production vehicle contractor inspection. The first completed production vehicle shall be road tested and inspected by the contractor as specified in table VI to determine conformance to contract and specifications. After inspection, the contractor shall submit the acceptable vehicle (and all inspection records and certifications) to the responsible Government inspection element at contractor plant for preliminary examination.

4.5.3 Preliminary examination. The responsible Government inspection element will conduct a preliminary examination, as specified in 4.6.1.3 and 4.6.1.4 of the first completed production vehicle.

4.5.4 Provisional inspection. Provisional inspection of the first completed production vehicle will be conducted jointly by representatives of TACOM Product Assurance Directorate and the responsible Government inspection element. Contractor shall provide any required assistance. Testing will be as specified in 4.6.2.2. Contractor shall make available his inspection plan, records, and certifications pertinent to the vehicle and components.

4.5.5 Repair of defects. Defects found as a result of foregoing inspections shall be corrected by the contractor at no cost to the Government. Failure of the contractor to promptly correct defects shall be cause for suspension of acceptance of vehicles until corrective action has been accomplished and approved by the Government.

4.5.6 Vehicle disposition. On completion of first production vehicle inspection, vehicle shall remain at manufacturing facility as a production sample and shall be the last vehicle shipped on contract. Vehicle may be released sooner at the discretion of the Government. Contractor shall service and maintain vehicle during this period.

4.5.7 Final approval and acceptance. Final approval and acceptance of the first production vehicle shall be withheld until first article initial production tests are accepted (see 4.4).

4.6 Quality conformance inspection.

4.6.1 Quality conformance inspection. Each production vehicle shall be subjected to a complete final inspection by the contractor as depicted in table VI and as described in 4.6.1.1 and 4.6.1.2.

4.6.1.1 Quality conformance examination. Each vehicle shall be inspected by the contractor for the characteristics/defects listed in table VI utilizing the Final Inspection Record (FIR) or its equivalent applicable to this vehicle to verify conformance of produced end items to the requirements of this specification. The AQLs depicted in table VII and VIII are listed for sampling purposes in the event sampling is authorized by the Government.

4.6.1.2 Quality conformance testing. Each vehicle after final inspection shall be operated for a distance of not less than 10 miles (16 km) by the contractor. Vehicle shall be completely assembled and serviced prior to road test. Performance of vehicles shall be demonstrated without payload

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at the place of manufacture on a smooth, relatively level, hard surfaced road to determine compliance to table VI.

4.6.1.3 Examination or test failure. If any vehicle fails to pass any examinations or tests (4.6.1.1 and 4.6.1.2) specified herein, the Government shall withold acceptance until evidence has been provided by the contractor that corrective action has been taken to correct all disclosed deficiencies.

4.6.2 Quality conformance inspection - sampling.

4.6.2.1 Lot size. An inspection lot shall consist of vehicles of a specific model, of one type, from an identifiable production period (i.e., one day's production), submitted at one time for inspection.

4.6.2.2 Sampling for quality conformance examination. Contractor shall 100 percent inspect first 20 vehicles, excluding first article and initial production samples, to establish process average (see 6.4). Thereafter, samples shall be selected from each lot in accordance with level II, MIL-STD-105.

4.6.2.3 Acceptable quality level. Visual, dimensional, and primary functional examination shall consist of examination of vehicle for conformance to applicable drawings and specifications. Examination shall be performed against classification of defects and with acceptable quality levels (AQL's) specified in Tables VII and VIII. Examination shall be performed during all phases of manufacturing and subsequent to road tests.

4.6.2.4 Classification of defects . For examination purposes, defects shall be classifed as specified in Table VII and VIII.

4.6.2.5 Unclassified defects. Defects considered to be departures from good workmanship but having no bearing on function, safety, interchangeability, or life shall be noted in writing. Such defects recurring, in five consecutive lots, or ten lots or more within a 30 day period, will be added to the minor classification of defects with no increase in AQL (see 6.5).

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TABLE VI. Classification of inspection.

Title	Requirement	First article preproduction	Initial production	Acceptance	Control	Comparison
Reliability	3.2.1		4.4.2			
Maintainability	3.2.2		4.4.3			
Physical configuration	3.4.1.1	4.7.1.1	4.7.1.1			
Track	3.4.1.2	4.7.1.2	4.7.1.2			
Curb weight	3.4.1.3	4.7.1.3	4.7.1.3			
Payload	3.4.1.4		4.7.1.4			4.7.1.4
Gross vehicle weight (GVW)	3.4.1.5	4.7.1.5	4.7.1.5			
Ground pressure	3.4.1.6	4.7.1.6	4.7.1.6			
Cargo space	3.4.1.7	4.7.1.7	4.7.1.7			4.7.1.7
Roof hatch	3.4.1.8	4.7.1.8	4.7.1.8	4.7.1.8	4.7.1.8	
Ground clearance	3.4.1.9	4.7.1.9	4.7.1.9	4.7.1.9		4.7.1.9
Mounts, mounting devices and brackets	3.4.1.10	4.7.1.10	4.7.1.10	4.7.1.10		4.7.1.10
Crew and cargo, protection	3.4.1.11	4.7.1.11	4.7.1.11			
Engine/fuel	3.4.1.12	4.7.1.12				
Transmission	3.4.1.13	4.7.1.13				
Lubricants	3.4.1.14	4.7.1.14				
Electrical system	3.4.1.15	4.7.1.15	4.7.1.15			
Electircal connectors	3.4.1.15.1	4.7.1.15	4.7.1.15			
Electrical circuits	3.4.1.15.2	4.7.1.15	4.7.1.15			
Batteries	3.4.1.15.3	4.7.1.15.1	4.7.1.15.1			
Lighting and blackout	3.4.1.15.4	4.7.1.15.2	4.7.1.15.2			4.7.1.15.2
Trailer lighting receptacle	3.4.1.15.5	4.7.1.15.3	4.7.1.15.3			
Intercommunications device/signal	3.4.1.16	4.7.1.16	4.7.1.16	4.7.1.16		
Accessories and equipment	3.4.1.17	4.7.1.17	4.7.1.17	4.7.1.17		
Personnel heater	3.4.1.18.1	4.7.1.18.1	4.7.1.18.1			
Engine heater	3.4.1.18.2	4.7.1.18.2	4.7.1.18.2			
Battery heater	3.4.1.18.3	4.7.1.18.3	4.7.2.28.3			
Windshield wipers and washers	3.4.1.19	4.7.1.19	4.7.1.19	4.7.1.19		

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TABLE VI. Classification of inspection.- Continued

Title	Requirement	First article preproduction	Initial production	Acceptance	Control	Comparison
Bilge pump(s)	3.4.1.20	4.7.1.20	4.7.1.20	4.7.1.20	4.7.1.20	4.7.1.20
Roof and roof rack	3.4.1.21	4.7.1.21	4.7.1.21	4.7.1.21		
Power winch	3.4.1.22	4.7.1.22	4.7.1.22	4.7.1.22		
Chemical biological (CB) clothing storage container(s)	3.4.1.23	4.7.1.23	4.7.1.23			
Speed	3.5.1	4.8.1	4.8.1	4.8.1	4.8.1	4.8.1
Terrain mix	3.5.2		4.8.2			4.8.2
Winter operation	3.5.3		4.8.3			4.8.3
Spring break-up and summer operation	3.5.4		4.8.4			4.8.4
Gradeability	3.5.5		4.8.5			4.8.5
Side slopes	3.5.6		4.8.6			4.8.6
Acceleration	3.5.7		4.8.7	4.8.7	4.8.7	4.8.7
Service brakes	3.5.8	4.8.8	4.8.8	4.8.8	4.8.8	4.8.8
Park brake	3.5.9	4.8.9	4.8.9	4.8.9	4.8.9	4.8.9
Turning ability	3.5.10	4.8.10	4.8.10			4.7.10
Vertical step	3.5.11		4.8.11			4.8.11
Standing vegetation	3.5.12		4.8.12			
Floatability	3.5.13		4.8.13			4.8.13
Skijoring	3.5.1.4		4.8.14			4.8.14
Towing	3.5.15		4.8.15			4.8.15
Range	3.5.16		4.8.16			4.8.16
Low temperature	3.5.17	4.8.17	4.8.17			
Power package cooling	3.5.18	4.8.18	4.8.18			4.8.18
Electromagnetic interference	3.6	4.9				4.8

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Table VII. Classification of major defects - AQL 25 defects/100 units.

Defect	Component	Defect	Method
101	Engine	Improper installation, unusual noise, leaks, malfunctions	Visual, functional
102	Transmission	Improper shifting, unusual noise, leaks, malfunctions	Visual, functional
103	Drive Train	Improper installation, unusual noise, malfunction	Visual, functional
104	Electrical system and components	Improper voltage, malfunction	Visual, functional
105	Track	Improper installation, damage	Visual, functional
106	Lubrication system components (engine)	Improper lubrication, damage, leakage	Visual, functional
107	Fuel tanks and system	Leakage, damage,cleanliness	Visual, functional
108	Suspension	Damage	Visual, functional
109	Exhaust system	Improper installation, leaks, excessive noise	Visual, functional
110	Instrumentation switches	Malfunction, location, damage	Visual, functional
111	Floatability	Excessive water leakage, malfunction	Visual, functional
112	Bilge pump	Malfunction	Visual, functional
113	Turnability	Nonconformance	Functional
114	Mounts, mounting devices, brackets	Improper location, missing	Visual
115	Roof rack	Improper location, ease of operation, missing	Visual, functional
116	Accessories	Missing, malfunction	Visual, functional
117	Engine heater	Malfunction	Visual, functional
118	Battery heater	Malfunction, missing	Visual, functional

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TABLE VIII. Classification of minor defects. - AQL 150 defects/100 units.

Defect Number	Component Assembly	Defect	Method
201	Coolant	Low or improper mixture levels	Visual, gage
202	Lubricants	Levels and proper types	Visual
203	Controls	Malfunction, adjustments, damage	Visual, functional
204	Roof hatches	Improper size and fit	Visual, functional
205	Paint	Improper color and application	Visual
206	Rustproof	Improper location or size, incomplete, data missing	Visual
208	Lube fittings	Improper installation, defective, missing	Visual
209	Body, doors, tiedowns, lifting device	Improper fit, defective weldment, inadequate seals	Visual,functional
210	Wiring or tubing	Improper assembly or installation, improper protection, defective	Visual

4.6.3 Quality conformance test - Sampling.

4.6.3.1 Acceptance test.

4.6.3.1.1 Sampling for acceptance testing. Samples for acceptance testing shall be selected in accordance with level S-3 of MIL-STD-105.

4.6.3.1.2 Acceptance testing. Each sample selected in accordance with 4.6.3.1.1 shall be examined as specified in 4.6.2, operated a distance of 50 miles (80.5 km) or more by the contractor and subjected to the acceptance tests of Table VII. except for equipment and component installation and servicing performed by contractor as a final processing of vehicles for acceptance. Vehicles shall be completely assembled and serviced with fuel, lubricants, and fluids as specified in Table II.

4.6.3.1.3 Failure. Failure to pass any acceptance test specified herein shall be cause for the Government to withhold acceptance until evidence of corrective action taken has been provided by the contractor and approved by the Government.

4.6.4 Control tests.

4.6.4.1 Sampling for control testing. Control test samples shall be selected at the rate of one of each 20 units produced. Not more than two nor less than one shall be tested in any 30 day period.

4.6.4.2 Control testing. Vehicles selected in accordance with 4.6.4.1 shall be examined for defects specified in Tables VII and VIII and subsequently subjected to the 50 mile test of 4.6.4.3 and tests specified in Table VI.

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4.6.4.3 Fifty mile test. Vehicle shall be loaded with full or simulated payload and operated for a distance of 50 miles over relatively smooth, hard-surfaced roads. During and subsequent to the road test, vehicle shall meet the requirements specified for control tests in Table VI.

4.6.4.4 Failure. Failure to pass any control examination test may be cause for the Government to refuse to accept subsequent lots until proven to the satisfaction of the Government that any deficiency has been corrected.

4.6.5 Comparison tests.

4.6.5.1 Sampling for comparison testing. Vehicles will be selected on a spot check basis in a quantity and at a time specified by the Government.

4.6.5.2 Comparison tests. Government may select vehicles at any time during the contract production period and subject them to Government proving ground testing. Sample vehicle shall be subjected to the comparison tests specified in Table VI, examined as specified in Tables VII and VIII and subjected to the 2,000 mile (3220 km) road test specified in Table IX. Samples shall not include any vehicles previously tested.

TABLE IX. Two thousand mile (3220 km) comparison test, combat loaded.

Terrain	Mileage (km)	% of total miles
Improved Roads	200 (322)	10
Unimproved Roads	600 (966)	30
Cross-Country	1200 (1931)	60

4.6.5.3 Failure. Failure of any vehicle to comply with the requirements specified herein or major vehicle defects in workmanship or material that occur during or resulting from the tests, may cause the Government to discontinue acceptance vehicles until evidence has been provided by the manufacturer that the defects have been eliminated.

4.7 Method of Inspection.

4.7.1 Physical characteristics.

4.7.1.1 Physical configuration examination. To determine conformance to 3.4.1.1, vehicle shall be examined for single or articulated two-unit configuration.

4.7.1.2 Track examination. To determine conformance to 3.4.12, vehicle shall be examined for one type track year-round service on all terrain and for damage after undergoing performance testing of 4.6.2.

4.7.1.3 Curb weight examination. To determine conformance to 3.4.1.3, vehicle shall be weighed for conformance and examined for attachments, accessories, equipment, and a full complement of fuels, lubricants, and coolant.

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4.7.1.4 Payload examination. To determine conformance to 3.4.1.4, vehicle payload shall be weighed for maximum payload requirements and examined for distribution of load over the vehicle body load space area.

4.7.1.5 Gross vehicle weight (GVW). To determine conformance to 3.4.1.5, vehicle shall be weighed for conformance and examined for proper payload.

4.7.1.6 Ground pressure examination. To determine conformance to 3.4.1.6, vehicle at GVW shall be examined and tested for maximum pressure.

4.7.1.7 Cargo space examination. To determine conformance to 3.4.1.7, vehicle shall be examined for cargo space enclosure and measured for minimum dimensional requirements.

4.7.1.8 Roof hatch examinations. To determine conformance to 3.4.1.8, vehicle shall be examined for a minimum roof hatch size opening and for a driver compartment roof hatch and cover.

4.7.1.9 Ground clearance examination. To determine conformance to 3.4.1.9, vehicle shall be examined on a hard, flat surface for minimum ground clearance.

4.7.1.10 Mounts, mounting devices, and bracket examination. To determine conformance to 3.4.1.10, vehicle shall be examined for accepting mounts, mounting devices, and brackets for attachments, accessories, and equipment specified herein.

4.7.1.11 Crew and cargo protection examination. To determine conformance to 3.4.1.11, vehicle crew and cargo compartment enclosure(s)shall be examined for protection from exposure to liquid chemical agent contamination.

4.7.1.12 Engine/fuel examination. To determine conformance to 3.4.1.12, vehicle shall be operated and examined for a fuel-efficient diesel engine capable of operating on fuel conforming to all grades of VV-F-800.

4.7.1.13 Transmission examination. To determine conformance to 3.4.1.13, vehicle shall be examined for automatic transmission.

4.7.1.14 Lubrication examination. To determine conformance to 3.4.1.14, vehicle shall be examined for lubrication chart specifying interchangeable military lubricants and shall conform to initial fills of lubricants specified herein and in Table II.

4.7.1.15 Electrical system examination. To determine conformance to 3.4.1.15 through 3.4.1.15.2, during vehicle operation the complete electrical system shall be operated and observed for functional requirements.

4.7.1.15.1 Batteries examination. To determine conformance to 3.4.1.15.3, vehicle batteries shall be examined for conformance to applicable documents and proper identification.

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4.7.1.15.2 Lighting and blackout examination. To determine conformance to 3.4.1.15.4, vehicle shall be examined for blackout window shades and tested for spectral emission of lamps.

4.7.1.15.3 Trailer lighting receptacle examination. To determine conformance to 3.4.1.15.5, trailer lighting receptacle shall be examined for proper terminal connections.

4.7.1.16 Intercommunications device/signal examination. To determine conformance to 3.4.1.16, vehicle shall be examined for a device to signal driver from cargo space.

4.7.1.17 Accessories and equipment examinations. To determine conformance to 3.4.1.17, vehicle shall be examined for standard accessories and equipment.

4.7.1.18 Heaters.

4.7.1.18.1 Personnel heater test. To determine conformance to 3.4.1.18.1, personnel heater vehicle shall be operated and tested for sufficient capability and capacity at waist level in driver area for sufficient defrosting of windshield and windows, and for maintaining cargo space reference temperature.

4.7.1.18.2 Engine heater test. To determine conformance to 3.4.1.18.2, vehicle engine heater shall be operated and tested for maintaining proper engine starting temperature.

4.7.1.18.3 Battery heater test. To determine conformance to 3.4.1.18.3, battery heater shall be operated and tested for maintaining a battery temperature able to accept a full charge when engine is operating.

4.7.1.19 Windshield wiper and washer examination. To determine conformance to 3.4.1.19, windshield wipers and washers shall be operated and examined for manual operation and conformance to Requirement 104 of MIL-STD-1180.

4.7.1.20 Bilge pump test. To determine conformance to 3.4.1.20, bilge pump shall be operated and tested for performance requirement.

4.7.1.21 Roof and roof rack examination. To determine conformance to 3.4.1.21, roof shall be examined for an anti-skid coating and tested for support of a 250 pound man. Roof rack shall be examined for proper strapping of skis and snowshoes to rack.

4.7.1.22 Power winch examnination. To determine conformance to 3.4.1.22, vehicle power winch shall be operated and examined for proper operation and performance requirements.

4.7.1.23 Chemical biological (CB) clothing storage container(s). To determine conformance to 3.4.1.23, vehicle shall be examined for storage container(s), CB protective clothing, and minimum size requirements.

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4.8 Performance tests. To determine conformance to 3.5, vehicle shall be tested with the standard service product specified in Table III.

4.8.1 Speed test. To determine conformance to 3.5.1, vehicle at GVW shall be loaded and operated maintaining top end speed and low speed for 30 minutes without steering difficulties, overheating, or malfunction of any component.

4.8.2 Terrain mix test. To determine conformance to 3.5.2, vehicle at GVW shall be operated over various types of terrain (see 3.2.1.1).

4.8.3 Winter operations test. To determine conformance to 3.5.3, vehicle at GVW shall be operated on crosscountry terrain, over snow-covered roads, and up to 6 feet of snow. Vehicle shall tow two 10-men infantry squads on skis while negotiating the terrain mix of 3.5.2.

4.8.4 Spring break-up and summer operation test. To determine conformance to 3.5.4, vehicle at GVW shall be operated over muskeg bogs, and muddy, gravel, or stone-covered roads. Vehicle shall be operated over the terrain mix of 3.5.2 during spring break-up and summer while traversing a minimum of 4.4 miles (7.0 km) in 1-hour.

4.8.5 Gradeability test. To determine conformance to 3.5.5, vehicle at GVW shall be operated ascending and descending forward and backward on hard, dry longitudinal grades up to 60 percent at a speed not less than 2 (mph)3.22(kmph). Vehicle shall be stopped and held motionless on a 60 percent grade, restarted, and continued ascending and descending the 60 percent grade. Vehicle shall be operated at not less than 2 mph (3.22 kmph) over a 30 percent longitudinal grade ascending and descending over 1.5 feet (0.46 m) of loose, dry snow.

4.8.6 Side slope test. To determine conformance to 3.5.6, vehicle at GVW shall be operated over a 40 percent hard, dry side slope without slipping sideways, (both left and right sides up slope) and steering shall be responsive and effective. Vehicle shall be stopped and held motionless on the 40 percent side slope, restarted, and shall continue traversing the 40 percent side slope. Vehicle shall be operated over a 30 percent side slope covered with 1.5 feet (0.46m) of loose, dry snow.

4.8.7 Acceleration test. To determine conformance to 3.5.7, vehicle at GVW shall be operated on dry, hard level roads from 0 to 25 mph (0 to 40.23 kmph) in less than 20 seconds.

4.8.8 Service brake test. To determine conformance to 3.5.8, vehicle brakes at GVW shall be applied while ascending and descending a 60 percent grade and shall stop at 50 feet (15.24m) or less from point of brake application, when traveling at a speed of 20 mph and towing a fully loaded 1 1/2 ton trailer. A maximum of 100 pounds force (444.8N) shall be applied on the brake pedal to achieve performance.

4.8.9 Park brake test. To determine conformance to 3.5.9, vehicle at GVW shall be parked up or down a 40 percent grade and held motionless with park brake applied.

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4.8.10 Turning ability test. To determine conformance to 3.5.10, vehicle shall be operated, turned to the right and left full 360 degree turns, and observed for performance.

4.8.11 Vertical step test. To determine conformance to 3.5.11, vehicle shall be operated over vertical obstacles of specified height.

4.8.12 Standing vegetation test. To determine conformance to 3.5.12, vehicle shall be operated over light and medium standing vegetation of specified height.

4.8.13 Floatability test. To determine conformance to 3.5.13, vehicle shall be driven into water of specified depth without special equipment and meeting performance requirements after floatation. Vehicle shall be driven in and out, forward and backward, from amd to a 20 percent dirt slope.

4.8.14 Skijoring test. To determine conformance to 3.5.14, vehicle shall be operated towing two 10-men squads on skis over specified terrain.

4.8.15 Towing test. To determine conformance to 3.5.15, vehicle shall be operated towing specified equipment and vehicles over cross-country terrain covered with a minimum of 1.5 feet (0.46 m) of loose, dry snow.

4.8.16 Range test. To determine conformance to 3.5.16, vehicle at GVW shall be operated 180 miles over improved roads with on-board permanent fuel tanks at speeds as specified.

4.8.17 Low temperature test. To determine conformance to 3.5.17, vehicle shall be stored and operated at temperatures specified.

4.8.18 Power package cooling test. To determine conformance to 3.5.18, vehicle shall be operated under specified conditions, maintaining specified steady state engine oil and transmission oil temperatures.

4.9 Electromagnetic interference tests. Vehicles electrical, electronic, and electromechanical equipment shall be subjected to test procedures specified in MIL-STD-462.

5. PACKAGING.

5.1 Processing and marking. Processing and marking shall be in accordance with the specific Equipment Preservation Data Sheet for the desired level of protection specifeid (see 6.2).

6. NOTES

6.1 Intended use. Vehicle covered by this specification is intended for use by the United States Army to carry selected items of existence equipment, ammunition, and supplies for use by small units during military operations. Vehicle is intended for use under the extreme or unusual conditions of climate, weather, and terrain in any season, northern and mountainous regions, and Alaskan and European terrain.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Basic issue items, pioneer tools, and other on-vehicle equipment provided with vehicle, shall be identified.
c. Painting as required (see 3.7).
d. Rustproofing as required (see 3.8).
e. Mounts, mounting devices, and brackets (see 3.4.1.10).
f. Quality of first article initial production test vehicles (see 4.4.1).
g. Winterization kit, manufacturer - furnished and installed (see 3.4.1.18.1).
h. Location for delivery of (IPT) Intial Production Test vehicles (see 4.4.1).
i. Shipment, level of protection (see 5.1)

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests, and approval of the first article (see 3.1).

6.4 Process average. Sampling may be initiated if the process average value for the first 20 vehicles inspected is less than the AQL specified in the classification of defects for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

6.5 Deficiency sheet. The Government inspector shall verify that a thorough inspection of each vehicle is performed by the contractor for the listed characteristics and any departures from good workmanship. The Government inspector shall assure that all defects encountered during inspection are enumerated on deficiency sheet for the vehicle. Defects noted on a deficiency sheet shall contain sufficient description to enable the Government inspector and the contractor representative to classify the defects in accordance with the classification (see Tables VII and VIII) and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring defects.

6.5.1 Recurring major defects. A major defect is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or equivalent) reflect such a condition. Recurring major defects shall be cause for the entire lot or lots to be inspected for recurring defects.

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6.5.2 Recurring minor defects. A minor defect is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor defects shall be cause for the entire lot or lots to be inspected for recurring defects.

6.6 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.3).

6.7 Changes from previous issue. Asterisks are not used in this revision due to the extensiveness of the changes, to identify changes with respect to the previous issue.

6.8 Electromagnetic compatibility. The contracting officer should submit a copy of the reports required (see 4.9) to Commander, US Army Communications Research and Development Command, ATTN: DRDCO-CM Fort Monmouth, NJ 07703 for approval.

6.9 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreements as set forth in NATO STANAG 4074. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization offices so that appropriate action may be taken respecting the international agreement concerned.

6.10 Safety precautions. Caution should be used in handling carbon dioxide extinguisher cylinders. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

Custodians:
Army - AT

Preparing activity:
Army - AT

Project No. 2320-AC26

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-62467(AT)	Carrier Cargo, 1½ Ton, M973, Small Unit Support Vehicle (SUSV)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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